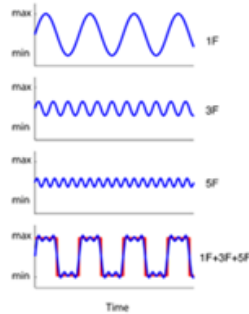


Data communications (Lab 1)

Question: Generating Square wave from composition of sine waves (Use Matlab). (5 marks)

Hints

A square wave consists of a fundamental sine wave and odd harmonics of the fundamental frequency. Generate a square wave with five odd frequencies. The composite square wave will be the summation of these sine waves. Use same amplitude for each of the sine waves. Use for loop. The result must show all the sine waves as well as the final square wave.



This MATLAB program demonstrates how a square wave can be constructed by adding multiple sine waves with odd frequencies. It also visualizes both individual components and the final waveform, making it useful for understanding signal synthesis.

```
clc;
clear;
close all;

% Time vector
t = 0:0.001:1;

% Fundamental frequency
f = 1;

% Number of odd harmonics
N = 5;

% Initialize square wave
square_wave = zeros(size(t));

figure;

% Loop for generating sine waves and summing them
for k = 1:N
    n = 2*k - 1; % odd numbers: 1,3,5,7,9

    % Generate sine wave
    sine_wave = sin(2*pi*n*f*t);
```

```

    % Add to composite signal
    square_wave = square_wave + sine_wave;

    % Plot individual sine waves
    subplot(N+1,1,k);
    plot(t, sine_wave);
    title(['Sine Wave with Frequency = ' num2str(n*f) ' Hz']);
end

% Plot final square wave approximation
subplot(N+1,1,N+1);
plot(t, square_wave, 'k', 'LineWidth', 1.5);
title('Approximate Square Wave (Sum of 5 Odd Harmonics)');
xlabel('Time');
ylabel('Amplitude');

```

A square wave can be approximated using a sum of sine waves with **odd harmonic frequencies**:

- Fundamental frequency $\rightarrow f$
- Odd harmonics $\rightarrow 3f, 5f, 7f, 9f$
- Total of 5 sine waves used
- All sine waves have the **same amplitude** (as required)

Code Explanation

1. Initialization

```

clc;
clear;
close all;

```

- `clc` $\rightarrow$ Clears command window
- `clear` $\rightarrow$ Removes all variables
- `close all` $\rightarrow$ Closes all figure windows

2. Time Vector

```

t = 0:0.001:1;

```

- Creates time values from **0 to 1 second**
- Step size = **0.001**
- Provides smooth waveform plotting

3. Define Parameters

```

f = 1;
N = 5;

```

- `f` $\rightarrow$ Fundamental frequency (1 Hz)
- `N` $\rightarrow$ Number of odd harmonics (5 waves)

4. Initialize Square Wave

```

square_wave = zeros(size(t));

```

- Creates an array of zeros

- Used to store the **sum of sine waves**

5. Create Figure

`figure;`

- Opens a new plotting window

6. Loop for Sine Wave Generation

```
for k = 1:N
    n = 2*k - 1;
```

Loop runs 5 times (k = 1 to 5)

`n = 2*k - 1` generates odd numbers:

- k=1 → n=1
- k=2 → n=3
- k=3 → n=5
- k=4 → n=7
- k=5 → n=9

7. Generate Each Sine Wave

```
sine_wave = sin(2*pi*n*f*t);
```

Standard sine function

Frequency = $n \times f$

Produces harmonic components

8. Add to Composite Signal

```
square_wave = square_wave + sine_wave;
```

9. Plot Individual Sine Waves

```
subplot(N+1,1,k);
plot(t, sine_wave);
title(['Sine Wave with Frequency = ' num2str(n*f) ' Hz']);
```

- Uses subplots to display each sine wave
- Total plots = 6 (5 sine + 1 square wave)

10. Plot Final Square Wave

```
subplot(N+1,1,N+1);
plot(t, square_wave, 'k', 'LineWidth', 1.5);
title('Approximate Square Wave (Sum of 5 Odd Harmonics)');
xlabel('Time');
ylabel('Amplitude');
```

- Displays the final combined waveform
- Appears similar to a square wave

